

Flock: Forecasting How Busy a Venue Will Be, and Counting Who Is There

Jayden Bansal

Flock Social LLC, Pennsylvania, USA
social@flockcorp.com

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Abstract

Flock is an app for planning nights out with friends, and its venue card answers one question: how busy will this place be when the group gets there? This paper studies both halves of that answer, forecasting a venue's busyness and counting the people in it. First, I ask whether a system that watches venues over time can beat the popular-times chart for one venue at one hour. Flock keeps each venue's weekly curve and corrects it with the venue's own recent live readings. On 4,183 live readings held out in time, the served number fell within one of five crowd levels 78.8% of the time and named the exact level 56.0% of the time, and its mean absolute error of 17.06 is 33.5% below the popular-times baseline. A gradient-boosted model of how an hour departs from the curve lowered the error from 31.48 to 29.42 on three cities it never saw. These forecasts predict busyness normalised per venue, not a number of people. The second study is Owl, a 718,550-parameter network that counts people on Flux, a venue sensor built around a 160×120 FLIR Lepton 3.5 thermal camera. It runs in 14–16 ms per frame on one core of a Raspberry Pi 5 and sends counts only. Trained on physics-based generated frames and six licensed real datasets, with no footage of people recorded for it, the shipped model counts 82–96% of frames exactly on five real test sets, where the heat-threshold rule it replaced managed 9–77%. The paper closes by setting out how Flux's counts could give the forecast an absolute scale and fresher readings, and what would be measured first.

Keywords: crowd prediction; popular times; nowcasting; gradient boosting; thermal imaging; people counting; on-device inference; privacy by design

Outline. Sections 2 to 7 cover the forecast: related work, preliminary findings, methods, results and discussion. Sections 8 to 18 cover Owl, the people counter. Section 19 joins the two. Appendices A and B hold supporting analyses for the forecast.

1 Introduction

1.1 Background and motivation

A group deciding where to go tonight wants one fact about each option: whether it will be busy when they arrive. Every consumer map gives the same answer, the venue's typical pattern for that hour of that weekday, averaged over a trailing window and drawn as a bar chart. Möhring et al. studied the Google form of it as a behavioural signal [11]. It is a strong prior. It is also blind, by construction, to the specific evening: the rain, the game two blocks away, the holiday, the week the place became popular.

I build Flock, an app for planning nights out with friends. Its venue card shows one integer from 0 to 100 and one of five words: *Quiet*, *Not Busy*, *Steady*, *Busy* and *Packed*. Whether that number is right is the difference between a plan that works and a group standing outside a packed bar. This project asks how much better than popular times that number can be, and what it takes.

1.2 Research question

On the venue-hours a production system actually serves, can a system that watches venues over time beat each venue’s own typical pattern, and by how much?

1.3 Hypotheses

- **H1.** Correcting the popular-times curve with the venue’s own recent live readings reduces error against held-out live readings, on both the point scale and the five-level scale.
- **H2.** A model that predicts how an hour departs from the curve, using context such as weather, events and calendars but no recent history, beats the curve on cities it has never seen.
- **H3.** The value of a live reading decays within hours, so the weight it deserves falls with its age.

1.4 Contributions

- Three measured properties of vendor typical-week curves that shape any system built on them: as forecasts they swing about twice as far as the readings they predict; a venue’s deviation from its curve fades within about three hours; and level metrics on lattice-valued labels carry a “grid floor” that favours doing nothing (Section 3).
- A serving path that adds a trailing per-venue offset and an age-weighted nowcast to the curve, with every weight fitted strictly before the rows that score it. It cuts the popular-times baseline’s MAE by 33.5% (Section 5.1).
- A delta model evaluated on whole held-out cities and on the served population only. It takes R^2 above zero at single-venue, single-hour granularity (Section 5.2).
- An evaluation apparatus that outlasts any one model: a ship gate against production as served, trainer-server parity tests, and a dataset that records its own provenance (Section 4).

1.5 Counting the room

Every number in Sections 3 to 7 is a busyness score normalised per venue: 100 means the busiest that venue gets, not a number of people. Sections 8 to 18 describe Owl, an on-device people counter for Flux, Flock’s venue sensor, which counts the people a thermal camera can see and sends only the count. Section 19 sets out how the two could be joined, so that the forecast is checked against real counts.

2 Related Work

Point-of-interest visit forecasting. The closest published task forecasts hourly visits to named points of interest. BysGNN [7] forecasts hourly visits to 400 POIs in each of five U.S. cities from SafeGraph data, whose hourly visits also underlie mobility network models of epidemic spread [3]. Against a

Historical Average baseline (the same weekday over the previous month) that its authors liken to the popular-times graph, BysGNN reduces MAE by 46% to 54% (Houston 8.860 to 4.095). Its more often quoted 4.34% to 6.71% margin is over the strongest competing graph model. It reads each venue’s own preceding weeks of hourly visits, a history a consumer app rarely holds. Its MAPE of 60% to 80% shows what precision the task admits; MAPE itself degenerates near zero [8].

Occupancy prediction and granularity. Bollenbach et al. [2] found that occupancy R^2 at a tourist site falls from 0.87 for the whole site to between -0.08 and -1.26 at individual entrances. Following Moore et al. [12], they read R^2 above 0.3 as weak, 0.5 as moderate and 0.7 as strong. My task sits at the finest granularity there is: one venue and one hour.

Beating the trivial baseline is not given. In a beach-occupancy study [1], a supervised random forest (MAE 98) and SARIMA (MAE 82) both lost to naive persistence (MAE 79). In emergency-department occupancy forecasting [16], a Temporal Fusion Transformer failed to improve significantly on ARIMA. In weekly foot-traffic forecasting, a simple weighted average beat an LSTM [9]. These results are why every candidate in this project must beat the number production serves on the same rows (Section 4.4).

3 Preliminary Findings

Before designing anything, I measured the popular-times baseline against provenance-verified live readings. Each finding below fixed one design decision.

3.1 The curve swings twice as far as the readings it predicts

I joined every verified live reading to its own venue’s curve value for the same weekday and hour. That gave 9,903 paired rows, on which the curve can be compared with what the room actually did. The gap is not a flat offset: it inverts with the level of the curve (Figure 1). In the hours a venue’s curve places in the bottom quarter of its day, observed busyness sits 15.60 points *above* the curve (1,162 rows). In the top quarter it sits 13.56 points *below* (5,746 rows). The middle quarters deviate by $+1.68$ and -4.99 .

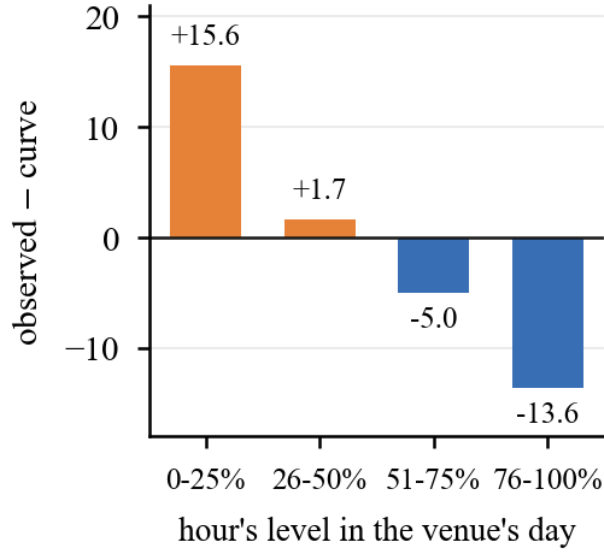


Figure 1. Observed minus curve, by the hour's level in the venue's own day (9,839 of the 9,903 paired rows).

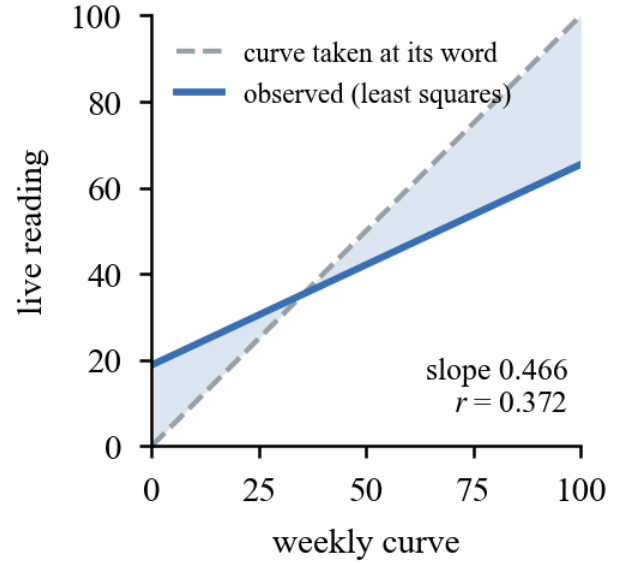


Figure 2. The least-squares fit of live readings on the curve (9,903 rows), against the identity line.

A least-squares fit gives $observed = 18.82 + 0.466 \times curve$, with $r = 0.372$ (Figure 2). The curve swings about 2.1 times as wide as the average reading at the same curve value, and explains about 14% of the variance in live readings. Individual readings are still more spread out than the curve (their standard deviation is about 1.25 times the curve's, from the same fit); what the curve overstates is how far the *expected* reading moves. This is regression toward the mean in a forecast [5, 6]. It also explained a mean deviation of about -11 that I had first read as a plain bias. **Design decision:** a correction must be able to move quiet hours up and busy hours down, which one global offset cannot do.

3.2 A venue's recent state persists, then fades

The vendor's live value is sticky: 80.9% of consecutive-hour readings of a venue are identical. A venue's deviation from its curve correlates 0.815 across one hour, 0.487 across two and 0.203 across three (Figure 3). Used as the prediction for the next hour, a reading one hour old lands within one crowd level 92.0% of the time and within 10 points 85.9% (3,165 readings, 2026-09-01 to 09-08). Nothing in a venue's typical week carries this information. **Design decision:** the latest reading should enter the served number directly, at a weight that decays on the same timescale.

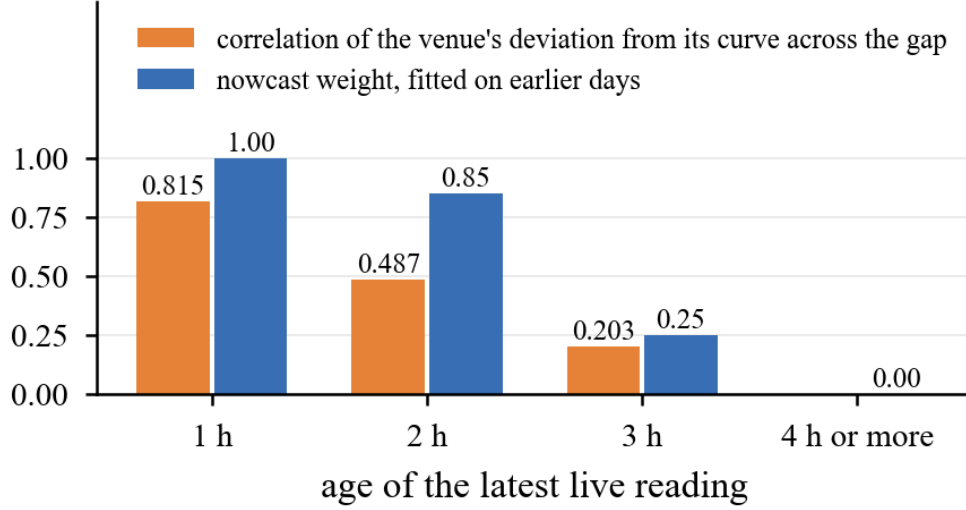


Figure 3. How a venue's deviation from its curve persists over time, against the nowcast weights later fitted on earlier days (Section 4.4). The fitted weights fall with age, though more slowly than the correlation at two hours.

3.3 The within-10 metric rewards the grid

Vendor labels lie on a lattice of multiples of 5, and the curve is a rounded average of lattice values, so it sits on the lattice by construction. A forward split shows what that does to the metric. I fitted each venue's offset on its earlier readings and scored its later ones (894 venues, 3,930 later readings) at offset weights λ from 0 to 1 (Figure 4).

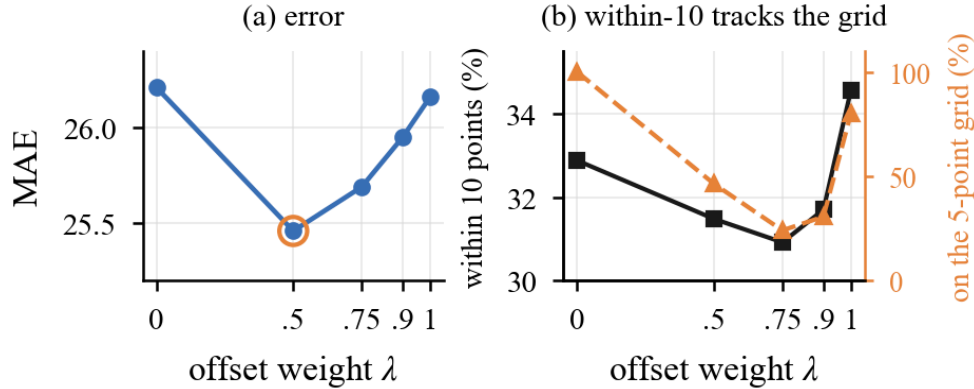


Figure 4. The forward offset test. (a) MAE is best at $\lambda = 0.5$. (b) The within-10 rate follows the share of predictions on the 5-point grid, not the error.

MAE is best at $\lambda = 0.5$ (25.46, against 26.21 with no offset). Within-10, however, is highest at the two weights whose predictions most often land on the grid: $\lambda = 1$ (34.55%, 79.9% on-grid) and $\lambda = 0$ (32.88%, 100% on-grid). The median error is 20 at every weight. On a lattice-valued dataset, within-10 partly measures grid coincidence. **Design decision:** where a gridded and a non-gridded predictor are compared on within-10, the comparison reports the on-grid rate or projects both to the lattice (Section 5.4); the within-10 margins in Table 3 are several times the grid effect measured here.

4 Methods

4.1 Definitions and problem statement

Busyness and crowd levels. For a venue v and an hour h , busyness $y \in [0, 100]$ is normalised per venue, so that 100 is the busiest that venue gets. The app publishes $\text{round}(\hat{y})$ and a word, cut at 20, 40, 70 and 85 into *Quiet*, *Not Busy*, *Steady*, *Busy* and *Packed*.

The weekly curve. Let $b(v, d, h)$ be the venue’s typical value for weekday d and hour h : the rounded mean of its weekly rows on the venue’s local clock, smoothed with its two neighbouring hours at weights 0.6, 0.2 and 0.2. This is the popular-times baseline.

Problem. Given a venue v , a target hour h , the curve b , and only the live readings of v taken strictly before h , produce $\hat{y}(v, h) \in \{0, \dots, 100\}$ and its crowd level, to minimise error against the live reading at h . The target hour’s own reading is never an input.

4.2 Data collection

Table 1. Data sources.

Source	Contribution
Vendor weekly forecasts	One typical week per venue: 168 rows on a 21-point lattice of multiples of 5. These rows define the curve.
Vendor realtime endpoint	One reading per venue-hour on a real date, labelled live or vendor forecast since 2026-08-15. About 1,379 Pennsylvania venues are read hourly, about 2,100 live readings a day; Miami venues are read at two fixed hours a day.
Weather	Temperature, humidity, wind, condition code and rain flag at the hour.
Events	Nearest ticketed event’s distance, type and attendance; counts within 2 km. For the October retrain, 627 professional and college games, refreshed daily.
Venue attributes	Category, rating, review count, price level, up to three place types, coordinates.
Calendars	Holiday, school break, holiday eve and special nights across 29 scopes.
Owner readings	A venue’s own live 0 to 100 report, at training weight 0.10.

Procedure. An hourly collector reads every tracked venue’s live value and stores one row per venue per hour, stamped with the venue’s local clock and a provenance label. Since 2026-09-25 the full sweep of about 1,379 venues completes in 8 min 42 s; before that, it had been stopping at its 50-minute budget. Weekly curves and context tables (weather, events, calendars) are refreshed on their own schedules.

Size. At the training run the corpus held 3,912,357 rows: 3,454,955 weekly and 457,402 realtime, across 34,784 venues in 34 cities, collected between 2026-03-10 and 2026-05-18. Realtime rows were scarce: 457,402 observations over 18,442 venues is 25 per venue across 168 weekly slots (26.3 after a 2026-09-06 duplicate-venue merge).

Integrity. Unique keys enforce one weekly row per (venue, weekday, hour) and one realtime row per date. Before they existed, I found and repaired three faults. The hour column was an array index in 3,536,360 rows (slot 18 held the venue’s midnight). There were 491,100 duplicate rows. And 63% of rows

carried an impossible month-season pair. No accuracy figure from before the repair is reported here. The 457,402 realtime rows collected before provenance was recorded cannot be separated into live readings and vendor forecasts, and stay marked *unknown*. “Live” in this report means the provenance-verified readings collected hourly since 2026-09-01.

4.3 Variables and controls

Table 2. Variables and controls.

Type	In this project
Independent variable	The serving configuration or model that produces the number: weekly curve, curve + offset, curve + offset + nowcast, rule engine, delta model v2.6.0, incumbent v2.5.0.
Dependent variables	Within one level, exact level, band MAE, MAE, R^2 , within 10 and within 15 points, bias.
Controlled: the rows	Every configuration is scored on the identical held-out readings, rebuilt through the production serving code.
Controlled: time	Every serving weight is fitted on 2026-09-01 to 09-05 and scored only on 09-06 to 09-08.
Controlled: geography	Three whole cities (Miami, Tokyo, Barcelona) never enter training; no venue appears on both sides of any comparison.
Controlled: population	Scored rows must satisfy the serving predicate (realtime, curve above zero), one function shared by trainer and evaluator.
Reference	A constant “Not Busy”, which shows how much of any level-agreement rate a do-nothing answer earns.

4.4 System design

Figure 5 shows the system. Three data sources feed two pipelines. The online pipeline serves every request. The offline pipeline trains a delta model, which can replace the online path only by passing a two-part ship gate.

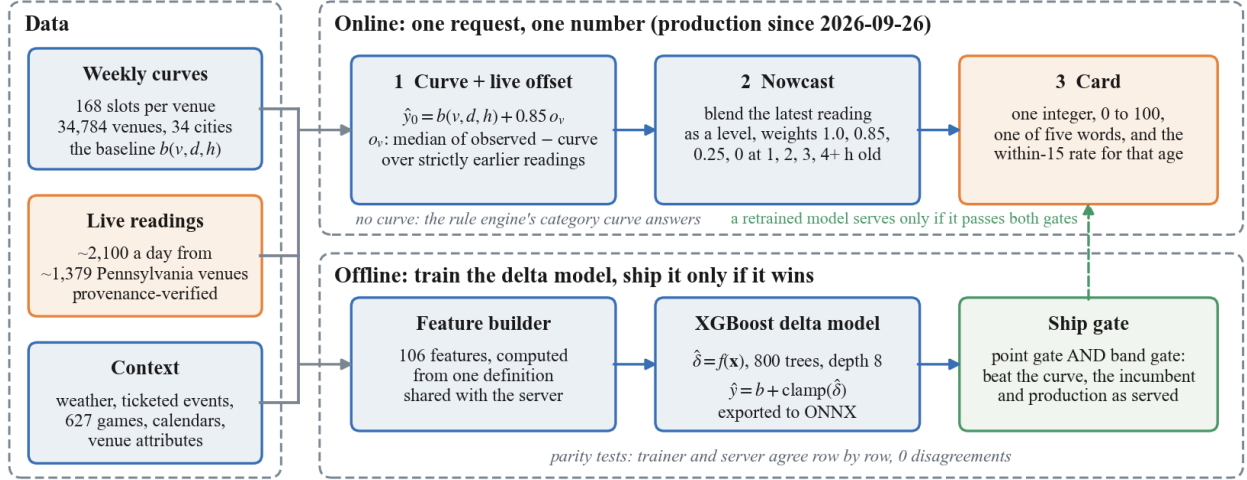


Figure 5. The system. One data column feeds an online serving path (in production since 2026-09-26) and an offline delta model, which serves only if it passes both halves of the ship gate.

Online correction. For each venue, let o_v be the median of (*observed* - *curve*) over the venue’s live readings taken strictly before the hour being scored. Let r_v be its most recent reading from an earlier hour, and a the age of that reading in hours. The served number is

$$\hat{y}_0(v, h) = b(v, d, h) + 0.85 \cdot o_v \quad (1)$$

$$\hat{y}(v, h) = (1 - w_a) \cdot \hat{y}_0(v, h) + w_a \cdot r_v \quad (2)$$

with $w_1 = 1.0$, $w_2 = 0.85$, $w_3 = 0.25$ and $w_a = 0$ for $a \geq 4$. The offset weight and all four blend weights were fitted by MAE on 2026-09-01 to 09-05 only. At an age of one hour the fitted weight is 1.0, so production then serves the last reading unchanged: the “repeat the last reading” baseline of Section 3.2 is the case the fit chose for fresh readings, and the offset takes over as readings age. The card also publishes a confidence: the within-15 rate measured for the number’s own reading age (Section 5.3).

Offline delta model. The model predicts how an hour departs from the curve, $\hat{\delta} = f(\mathbf{x})$, from 106 features:

- the category curve and a category-by-hour curve;
- venue attributes (13 categories, price, rating, reviews and 30 place types);
- calendar (hour, weekday, cyclic encodings, weekend, holiday, school break, month, season, daylight);
- weather against a monthly climate table;
- ticketed events;
- aggregates of in-app feedback.

Every column that would leak the label is excluded, following Kapoor and Narayanan’s taxonomy [10]. The learner is XGBoost [4]: 800 trees at depth 8, learning rate 0.01, subsample 0.9, column subsample 0.8, minimum child weight 7, L1 0.5, L2 1.0. The score is rebuilt by a fixed chain, clipped to $[0, 100]$ and

rounded:

$$\hat{y} = \text{round}(\text{clip}(b + 0.5 o_v + \text{clamp}(\hat{\delta}, \pm 50), 0, 100)). \quad (3)$$

On the geographic holdout no venue has earlier readings, so $o_v = 0$ there. The v2.6.0 ship gate scored the delta with a ± 30 clamp and without rounding (Table 5); Section 5.2 gives the model under Equation (3) exactly.

Ship gate. A candidate model must pass a point gate and a band gate.

- *Point gate:* on the held-out cities, beat the curve by 5 MAE or 0.10 R^2 , not regress MAE, and match the incumbent’s own within-10 on the same rows.
- *Band gate:* on at least 1,000 held-out live readings over 5 or more dates, it must:
 - beat production *as served* on within one level;
 - not be worse on within-10 (the date-block 95% lower bound above -1 point);
 - beat the weekly curve with a lower band MAE;
 - hold every city with 300 or more readings within 2 points.

The server fails closed: an artifact that did not pass cannot be loaded.

4.5 Evaluation protocol and statistics

- **Time holdout.** The served number is scored on 4,183 live readings from 2026-09-06 to 09-08 (Lehigh Valley 3,615, Miami 568), each rebuilt through the production serving code.
- **Geographic holdout.** The delta model is scored on the *gate slice*: the 67,249 realtime rows with a usable curve in the three held-out cities.
- **Cross-validation.** Leave-one-city-out over the 30 training cities, the block design recommended for spatially structured data [15].
- **Reproduction.** Every geographic-holdout number is recomputed from the saved model and holdout by scripts that pin seeds and thread counts, and those scripts also produce the intervals and the learner comparison (Tables 7 and 10).
- **Statistics.** Intervals are date-block bootstraps that resample whole observation dates, quoted wherever the record carries them. The per-reading rows behind Table 3 are not archived with this report, so its intervals are computed from the published rates and n : a Wilson interval for each rate, and for each difference an interval that treats the two arms as independent samples. Both arms are scored on the same readings, which makes that interval conservative; readings of one venue on one night are correlated, which cuts the other way, so each difference also states how far clustering would have to inflate the variance before its interval reached zero (Table 4). Level agreement is never reported without exact level and band MAE, because a constant can score as high on it (Section 5.1).

4.6 Reproducibility safeguards

Training pins the device, thread count and seed, so two runs on the same data produce bit-identical predictions [14]. The model is exported to ONNX and served in-process through ONNX Runtime [13]. Each feature family is computed by the trainer and the server from one definition, and a test requires them to agree row by row. All report zero disagreements:

- the neighbour arithmetic, on 8,006 readings;
- the live-reading features;
- the game features;
- 62 calendar, weather and event columns;
- the event collection, on 400 listings.

An offline replay used for every live score is proven by a test to publish exactly what the serving function publishes.

5 Results

5.1 H1: the served number on held-out live readings

Table 3. The served number and its comparators on 4,183 held-out live readings (2026-09-06 to 09-08; weights fitted on 09-01 to 09-05). Bold marks the best value in each column. Bias is the mean of predicted minus observed. Blank cells were not recorded for that arm. The last row was rehearsed and not shipped (Section 5.6).

Configuration	Within one level ↑	Exact level ↑	Within 10 ↑	Band MAE ↓	MAE ↓	Bias
Production: curve + offset + nowcast	78.8%	56.0%	53.8%	0.735	17.06	-0.98
Curve + offset, no nowcast	70.2%	33.5%	30.2%	1.061	24.45	-1.75
Weekly curve alone (popular times)	65.9%	31.5%	32.3%	1.128	25.66	+4.39
Rule engine	61.7%	22.8%		1.269	29.03	
Before: v2.6.0 as served (map on)	58.5%	32.2%	28.6%	1.321	30.61	+13.00
<i>Constant “Not Busy” (reference)</i>	78.8%	14.2%		1.212	26.82	
Retrain with live features (not shipped)	79.7%	43.1%	41.6%	0.817	18.96	+0.40

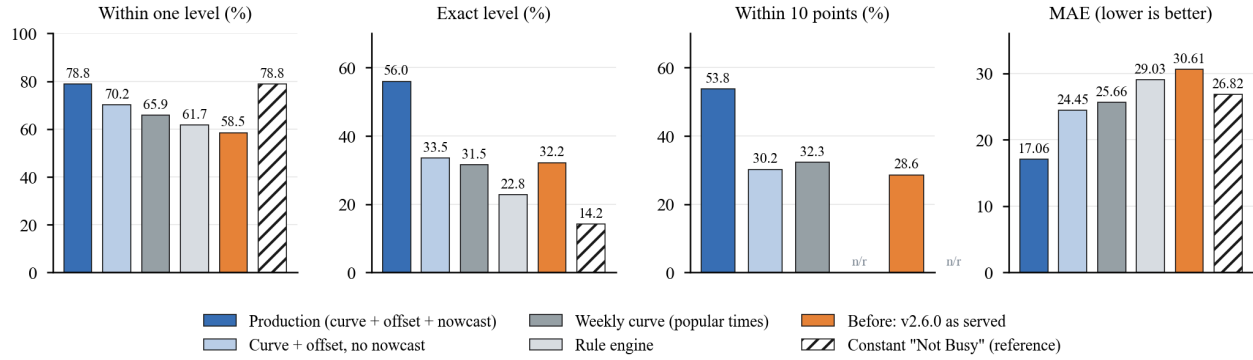


Figure 6. The served number against its comparators on 4,183 held-out live readings. n/r: not recorded for that arm.

Against the configuration it replaced, the served number gained 20.3 points within one level, 23.8 on the exact level and 25.2 within 10 points. MAE fell 44% and the bias fell from +13.0 to -0.98. Against the popular-times baseline, MAE is 33.5% lower and within-10 is 21.5 points higher. The constant row keeps the result honest. Real venue-hours are bimodal (37.7% *Quiet* and 16.5% *Packed* in September), and “Not Busy” sits within one level of three of the five levels. A constant therefore scores the same 78.8% within one level, but names the exact level only 14.2% of the time, with a band MAE of 1.212 against production’s 0.735.

Table 4. Production’s margin over each comparator on the same 4,183 readings, in points, with 95% intervals that treat the two arms as independent samples. The last column is the factor by which clustering would have to inflate the variance before the smallest margin in the row reached zero.

Production minus	Within one level	Exact level	Within 10	Inflation to reach zero
Weekly curve alone (popular times)	+12.9 (11.0 to 14.8)	+24.5 (22.4 to 26.6)	+21.5 (19.4 to 23.6)	46×
Curve + offset, no nowcast	+8.6 (6.7 to 10.5)	+22.5 (20.4 to 24.6)	+23.6 (21.5 to 25.7)	21×
Rule engine	+17.1 (15.2 to 19.0)	+33.2 (31.2 to 35.2)		79×
Before: v2.6.0 as served	+20.3 (18.4 to 22.2)	+23.8 (21.7 to 25.9)	+25.2 (23.2 to 27.2)	110×
Constant “Not Busy”	0.0 (-1.8 to 1.8)	+41.8 (40.0 to 43.6)		

How much clustering inflates the variance is itself uncertain. The two comparisons on record that carry both a date-block interval and a row-level one (Section 5.6) put it at about 5 and 10 times, but they rest on three dates. On the spring gate slice, date blocks inflated the variance of an MAE difference about 55 times (Section 5.2). At 55 times, every exact-level and within-10 margin in Table 4 still holds, and so does every margin over the rule engine and the old model; the within-one-level margins over popular

times (46 times) and over curve plus offset (21 times) would not. The October test, over 14 dates, settles which regime the live readings are in. The constant ties production on within one level, which is the point it was included to make, and loses by 41.8 points on the exact level. **H1 is supported.**

5.2 H2: the delta model on cities it never saw

Table 5. Point accuracy on the 67,249-row gate slice, scored as the v2.6.0 ship gate scored it: the predicted delta clamped at ± 30 and not rounded, with $\sigma_v = 0$ because no held-out venue has earlier readings (display map off).

Predictor	MAE ↓	R ² ↑	Within 10 ↑
Weekly curve alone	31.48	-0.075	19.2%
Incumbent v2.5.0	30.77	-0.043	19.3%
Delta model v2.6.0	29.42	+0.040	20.7%

The model improves on the curve by 2.06 MAE (6.5%) and 0.11 R², and on the incumbent by 1.35 MAE. R² crosses zero: this is the first version whose predictions carry more information than the mean of the labels. The gain holds in every held-out city (Figure 7): 2.26 MAE in Miami, 2.21 in Barcelona and 1.72 in Tokyo, where within-10 is 23.6%, 20.9% and 16.9%. Every number in this section reproduces exactly from the saved model and holdout (30 of 30 checks; the incumbent from its tagged commit).

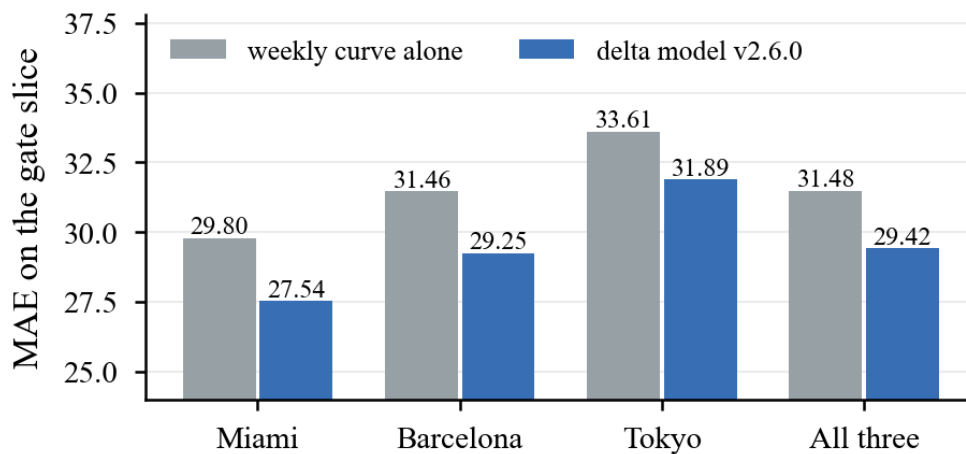


Figure 7. Delta model against the weekly curve on each held-out city of the gate slice.

On the same rows, measured on the five-level scale (Table 6), the model wins on every column: +2.6 points of exact level, +4.3 within one level, and 0.086 fewer levels of error on average.

Table 6. Level accuracy on the gate slice, with the delta clamped at ± 50 and not rounded, and a score of exactly 20 counted as *Not Busy* (display map off). Production counts 20 as *Quiet*; on that ladder the gains are +2.5 exact, +4.4 within one level and 0.089 fewer levels.

Predictor	Exact $\uparrow$	Within one $\uparrow$	Band MAE $\downarrow$	MAE $\downarrow$
Weekly curve	22.9%	57.7%	1.315	31.48
Delta model	25.5%	62.0%	1.230	29.35

Scored exactly as Equation (3) writes it, with the ± 50 clamp and rounding, the model reaches MAE 29.35, R^2 +0.042 and within-10 21.7%. Rounding onto the labels’ 5-point grid adds a point of within-10: the grid effect of Section 3.3 again.

Table 7. The delta model’s margin over the weekly curve on the gate slice, with 95% intervals from a paired two-way cluster bootstrap (2,000 resamples, stratified by city) that resamples venues and city $\times$ month $\times$ weekday blocks. Each metric is scored on the chain of its own table.

Held-out rows	MAE	Within 10	Exact level	Within one level
All three cities	-2.06 (-2.75 to -1.44)	+1.51 (0.44 to 2.65)	+2.61 (1.55 to 3.79)	+4.31 (2.93 to 5.80)
Miami	-2.26 (-3.50 to -1.13)	+1.87 (-0.01 to 3.93)	+2.62 (0.65 to 4.78)	+4.83 (2.39 to 7.70)
Barcelona	-2.21 (-3.22 to -1.27)	+2.54 (0.38 to 4.87)	+4.27 (2.19 to 6.55)	+3.68 (1.79 to 5.76)
Tokyo	-1.72 (-2.69 to -0.83)	+0.50 (-0.66 to 1.85)	+1.71 (0.46 to 3.23)	+3.98 (2.01 to 6.17)

The holdout carries no venue or date column, so both clusters had to be recovered. Venues come from their static attributes: the daylight feature is an exact function of latitude and month, which identifies 2,644 venues, each holding exactly one row per weekly slot. City $\times$ month $\times$ weekday is the finest date proxy the rows carry. The choice matters. The dependence is temporal: date blocks widen the MAE interval 7.4 times over treating rows as independent, where venues alone widen it 1.4 times. R^2 gains 0.115 (0.079 to 0.154), band MAE falls 0.086 (0.058 to 0.116), and against the incumbent v2.5.0 MAE falls 1.35 (1.00 to 1.72). Every margin excludes zero overall and in every city except within-10 in Miami and in Tokyo taken alone, which is why the claim for H2 rests on MAE and the level metrics.

Out of fold, where every training row is scored by a model that never saw its city, the 369,076 served realtime rows give MAE 27.54, R^2 0.127, within-10 22.8% and within-15 33.3%. Fifteen days forward of the training window, on 103,254 realtime rows across the 30 training cities, the model scores MAE 29.02 against 30.23 for the curve, and R^2 0.078 against 0.016. The gain is smallest in the dinner (17:00 to 20:00) and evening (21:00 to 23:00) hours, 0.96 and 0.87 MAE, which are the hours the app is used most.

Placed against prior work, the 6.5% comes with no recent history, where BysGNN’s 46% to 54% uses weeks of it [7]. An R^2 of +0.040 is above zero where the nearest peer-reviewed occupancy study sits between -0.08 and -1.26 [2]. It is also below the floor that study reads as “weak” [12]. **H2 is supported on held-out cities, with a caveat:** the model was trained on spring rows of unknown provenance. On verified live readings the spring-trained model loses to the untouched curve as it was served (Section 6.2). That is why production now serves the online correction instead.

5.3 H3: the value of a reading by its age

The served number's accuracy is set largely by how recently the venue was read (Figure 8). With a reading one hour old, 87.4% of served numbers land within 15 points. That falls to 62.8% at two hours and 43.5% at three, close to the 41.7% of curve and offset alone. In September, 41% of readings had one an hour earlier; the faster hourly sweep raises that share. **H3 is supported:** the decay in accuracy matches the decay in correlation measured in Section 3.2.

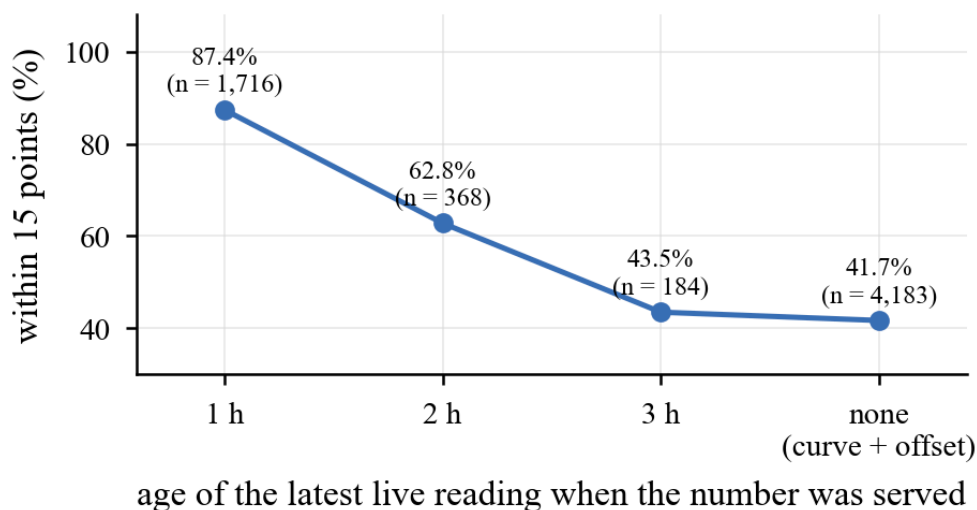


Figure 8. Accuracy by the age of the latest reading, on the held-out live days. The app publishes this rate as its confidence, per reading age.

5.4 What coverage is worth

A delta model cannot score a venue that has no curve, so I compared the model path with the rule engine that serves such venues. Both were given identical inputs on 5,025 verified live readings in Pennsylvania at venues that hold their own curve, with the events lookup disabled for both (Table 8, Figure 9).

Table 8. The two serving paths on the same 5,025 live readings (display map on, as served then).

Path	MAE ↓	Within 10 ↑	Within 15 ↑	Median error ↓	On-grid
Model path	28.22	31.7%	41.5%	21	41.5%
Rule engine	30.04	20.3%	29.2%	27	19.8%
Curve alone	26.61	30.0%	41.5%	20	100%

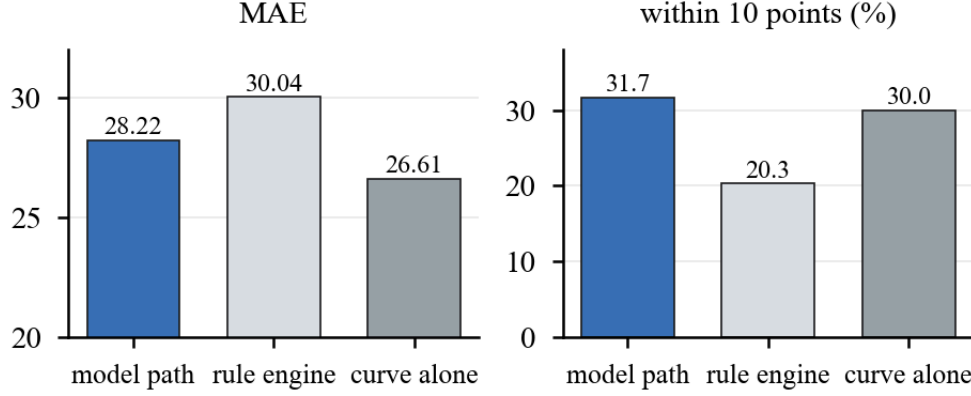


Figure 9. The model path and the rule engine on the same 5,025 live readings, with the curve alone for reference.

Coverage is worth 1.82 MAE, 11.4 points of within-10 and six points of median error. These are floors, since disabling events handicaps only the model arm. Grid-equalised, coverage is still worth 11.2 points of within-10, so the headline is not a lattice effect. The bare curve beats the model path on MAE by 1.61 while losing on within-10: on these bimodal labels the two metrics reward different predictions (Section 6.1).

5.5 Ablation experiments

Nowcast and offset. Table 3 separates the two corrections:

- *The offset* alone takes the curve from 65.9% to 70.2% within one level, from 31.5% to 33.5% exact, and from 25.66 to 24.45 MAE. It moves the bias from +4.39 to -1.75. Its within-10 falls from 32.3% to 30.2%: the grid penalty of Section 3.3 in miniature.
- *The nowcast* then does most of the work: within-10 rises from 30.2% to 53.8%, and the exact level from 33.5% to 56.0%.

Pulling the curve’s swing in directly. The obvious remedy for Section 3.1 is to rescale the curve toward the mean. I priced it on the 9,903 paired rows (Table 9, Figure 10) and rejected it.

Table 9. Corrections of the curve’s over-wide swing on 9,903 paired rows. The two per-venue rows are fitted and scored on the same readings, so they are upper bounds.

Predictor	MAE ↓	Within 10 ↑
Curve alone	26.86	31.6%
Curve + one global mean offset	25.88	28.9%
Global slope correction	25.49	21.8%
Per-venue intercept, half weight (in-sample)	23.18	35.9%
Slope + per-venue intercept (in-sample)	22.55	27.2%

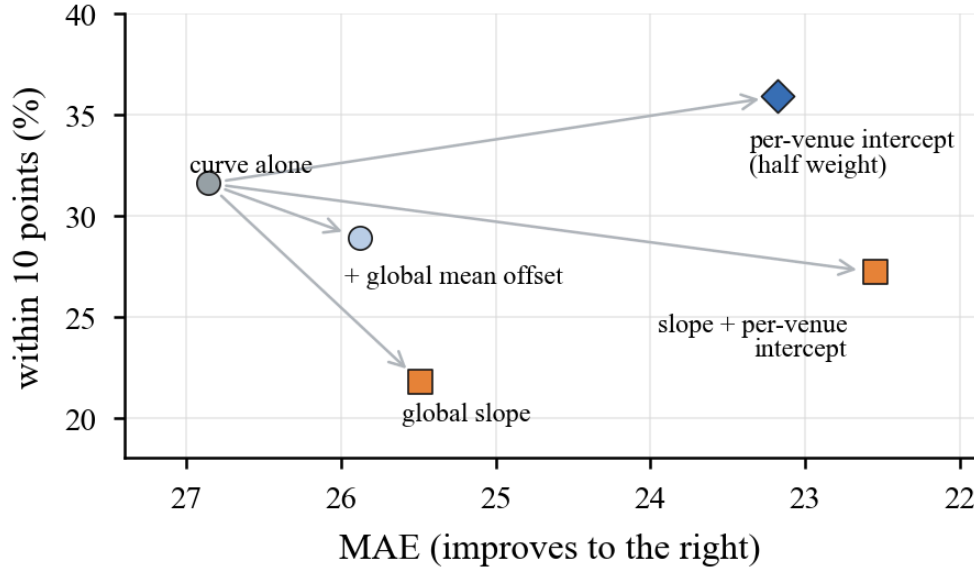


Figure 10. Pricing the corrections of Table 9. Corrections that narrow the spread (squares) buy MAE and spend within-10. Only the per-venue intercept, the offset of Equation (1), improves both.

Every correction that narrows the published spread buys MAE and spends within-10. A leave-one-out test over 688 venues (11,804 held-out readings) repeats the pattern out of sample. A per-venue slope buys 0.99 MAE over the per-venue intercept and costs 3.19 points of within-10, while the intercept itself buys 1.56 MAE against the bare curve (25.97 to 24.41) at a cost of 0.1 point of within-10. On 8,895 September live readings, the half-weight offset moves MAE from 25.90 to 24.53 and exact level from 32.1% to 38.0%, on five of five days and in both cities.

The display map. From 2026-08-28 to 09-26 a 41-knot monotone quantile map sat between rounding and the level cut. On the gate slice it bought 7.6 points of exact level and 7.4 points of within-10, and paid 3.2 MAE for them. On spring rows that looked like a fair trade. September's live readings reversed the verdict: the map cost 10.5 points of within one level (57.4% with it against 67.9% without, on 7,920 readings), and I removed it.

Volume against depth. A paired test held 17,322 rows out by both venue and date and scored them with two models. One was trained with the scored venues' own history, 304,000 more rows in total; the other had never seen those venues. The model with the extra history scored 27.63 MAE against 27.14, worse by 0.49. A two-head variant makes the same point from the other side:

- trained on the unknown-provenance pool, it beats the shipped model by 0.88 MAE (95% interval 0.51 to 1.27);
- held to the 4,223 verified labels available at the time, it loses by 1.59 MAE (interval 0.86 to 2.45).

The limit is the supply of verified labels per venue, not the shape of the model.

The learner. Nothing above says why XGBoost. To test that choice I held the features, the target and the score chain fixed and changed only the learner, in a rotation across the three held-out cities: each learner trained on two and was scored on the third (Table 10). Each fold gives a learner 135,000 to

165,000 rows from two cities, far fewer than the shipped model’s 1.93 million from thirty, so the shipped model appears for reference only. Neither boosted model was tuned, seeds and thread counts were pinned, and two runs gave identical predictions.

Table 10. Five learners with the same 106 features, target and score chain (Equation (3)), each trained on two held-out cities and scored on the third, pooled over the 67,249 gate rows. Bold marks the best learner in each column.

Learner	MAE ↓	R ² ↑	Within 10 ↑	Exact ↑	Within one ↑
XGBoost, the settings of Section 4.4	29.97	−0.000	21.4%	25.1%	60.7%
LightGBM	29.80	+0.010	21.7%	25.3%	61.1%
Ridge regression	30.18	+0.003	20.3%	24.1%	59.9%
Category-by-hour mean delta	31.61	−0.095	19.7%	23.3%	57.7%
Weekly curve alone	31.48	−0.075	19.2%	22.9%	57.7%
<i>Shipped v2.6.0 (thirty other cities)</i>	29.35	+0.042	21.7%	25.4%	62.0%

Against XGBoost, LightGBM is better by 0.18 MAE (0.07 to 0.29) and level with it on within-10; ridge regression is within 0.21 MAE (−0.02 to 0.49) but loses 1.09 points of within-10 (0.54 to 1.72) and 1.01 of the exact level; a category-by-hour average is no better than the curve. The evidence supports a gradient-boosted tree ensemble, not XGBoost in particular. The shipped model beats every learner here by at least 0.45 MAE: breadth across thirty cities is what helps a city with no history, a different claim from the one above, where more rows for the same venues did not help.

5.6 The retrain, rehearsed

A model trained with five live features on only 3,579 live readings reached 79.7% within one level, +21.2 points over v2.6.0 as served (95% interval 15.8 to 24.2). The features were the last reading’s deviation and age, the trailing offset and its count, and the curve one hour earlier; they carry 47% of its split gain. It still trailed production by 12.2 points of within-10 (interval −16.3 to −2.7), so the band gate refused it, exactly as designed.

6 Discussion

6.1 Interpretation

The venue’s own recent state is the lever. The closest published system halves its baseline’s error with recent observations [7]. Given the venue’s latest live reading, the served number cuts the curve’s error by 33.5% and raises within-10 from 32.3% to 53.8%. That is the clearest measurement in this project of what the next feature is worth.

The model’s gain is real, small and consistent. A 6.5% MAE reduction, and R² crossing zero at single-venue, single-hour granularity, is what the spring data allowed without any observation of the venue’s recent state. On the gate rows the model’s output correlates with the raw curve at 0.9638: the curve supplies the rhythm and most of the level, and the model supplies a correction.

Evidence density is the binding constraint. The spring corpus holds about 26 realtime rows per venue across 168 weekly slots. On 2026-09-06 only 9,339 rows across 1,228 venues carried verified live provenance, a median of six each. More rows of the same kind do not help (Section 5.5); the hourly live collection, about 2,100 readings a day, is the depth the problem asks for.

One number cannot sit near both modes. Of served venue-hours, 23.6% are at 5 or below and 22.1% at 90 or above. The labels have a standard deviation of 36.65, where the predictions have 21.56. MAE and within-10 give opposite instructions in that regime, at 0.16 to 0.36 MAE per point of within-10, and no calibration step after the model escapes the trade-off. A live reading is the first input that moved both metrics at once.

Who it helps. For a group choosing where to go, moving the exact level from 32.2% to 56.0% turns the card from a guess into an answer that is right more often than not. For a venue, a number people trust about the present is what can make a quiet hour worth visiting, and move some demand off the peak. For anyone building on vendor curves, two findings carry over without any of this code: a typical-week curve used as a forecast overstates how far the expected reading moves and should be judged as a forecast, and a within- k metric on lattice-valued labels should be read beside its on-grid rate. The numbers are venue-level aggregates from a vendor; no person's location enters them.

6.2 Limitations

- **Provenance.** The model results in Section 5.2 rest on spring rows of unknown provenance. On verified live readings, the spring-trained model loses to the untouched curve as served: MAE 30.61 against 25.66 on the held-out days, and 31.00 against 26.42 on all 7,920 September readings. With the map off it is only level (25.41 and 25.90).
- **A small comparator test.** The bake-off of Table 10 is untuned and confined to three cities of spring data. No classification or quantile formulation and no hyperparameter re-search was tried.
- **Latitude by another route.** Daylight hours is an exact function of latitude and month, so latitude re-enters the model through it, although coordinates were dropped as features.
- **Constant features.** Ten of the 106 features are constant: four season flags and the holiday flag (one spring of collection), four feedback aggregates (no verified feedback yet), and one event-type flag.
- **A short live window.** Section 5.1 rests on three dates in the Lehigh Valley and Miami, so its intervals span only three date blocks. The three were a Sunday, Labor Day and a Tuesday, so a holiday is one third of the test. The online correction went into production on those three dates, fewer than the five the band gate asks of a model artifact; it was not put through that gate, and the October test (Section 6.4) is the first to score it over more than three dates. *Packed* rooms are the one level the switches made worse: 53.0% within one level against 62.3% before, on the scored dates.
- **Attribution.** Part of the model's gain over the incumbent came from repairing the dataset rather than from modelling.

6.3 Future work

A retrain in mid-October will use roughly 90,000 to 95,000 live readings: about 60,000 to train, and the last 14 days, about 29,000 readings, held out. It will learn the live features rehearsed in Section 5.6 rather than blend them after the fact. It will be gated against production as it serves today, and shipped only if it wins. After that come a hyperparameter re-search, and fitting the output as a classification over the five levels or as quantiles.

6.4 A pre-registered test

Hypotheses H1 to H3 were stated after the analyses that test them (Appendix B). The October retrain is the first test in this project whose hypotheses, data split and decision rule were fixed before its data existed. They are fixed in code, not only in prose: the band gate’s thresholds and the 14-day time holdout were committed on 2026-09-25 and 2026-09-26, three weeks before the held-out window closes.

- **H4.** The retrain beats production as served on within one level over the held-out 14 days, with the date-block 95% lower bound above -1.0 point.
- **H5.** It is not worse on within 10, by the same bound.
- **H6.** It beats the weekly curve on within one level with a lower bound above zero, and its band MAE is no worse than the curve’s.
- **Guards.** At least 1,000 readings over at least 5 dates; production never saw the window; no city with 300 or more readings falls more than 2 points behind production; MAE rises by at most 1 point.
- **Analysis.** 2,000 date-block bootstrap resamples. The constant-answer reference is recorded beside every verdict.
- **Decision.** The retrain ships only if all of these hold. Otherwise production keeps serving, and the result is reported either way.

Table 11. What about 29,000 held-out readings can resolve: the smallest difference detectable at 80% power (two-sided, 5%), treating readings as independent and with the variance inflated four times, and the readings needed to detect a 1-point difference.

Metric (assumed rate)	Detectable, independent	Detectable, inflated 4×	Readings for 1 point
Within one level (79%)	0.95	1.90	26,000
Exact level (56%)	1.15	2.31	38,700
Within 10 (54%)	1.16	2.32	39,000

The arithmetic exposes a property of the gate. A 95% interval on a within-10 difference is ± 0.81 points at this sample size if readings were independent. With the 5 to 10 times inflation measured in Section 5.1 it is ± 1.8 to ± 2.6 points, so a retrain that exactly ties production on within 10 would see its lower

bound fall below -1 point and be refused. In effect H5 asks the retrain to win on within 10 by one to two points, not merely to match. I have kept it that strict on purpose: the rehearsal in Section 5.6 is the case it exists to stop.

7 Conclusions on the forecast

The number Flock serves today corrects the venue's curve with its recent live offset and a nowcast of its latest reading. On held-out live readings it lands within one crowd level 78.8% of the time, names the exact level 56.0% of the time, and comes within 10 points 53.8% of the time. The configuration it replaced scored 58.5%, 32.2% and 28.6%, and the served number's MAE is 33.5% below the popular-times baseline's. Beneath it, a gradient-boosted delta model beats the curve on whole held-out cities without any recent-history input.

Three findings should transfer to any system built on vendor typical-week curves:

- as forecasts, those curves swing about twice as far as the readings they predict;
- level metrics on lattice-valued labels carry a grid floor that favours doing nothing;
- the dataset that helps is not larger but deeper.

8 Counting people on a thermal camera

Flock is an app for planning nights out with friends, and one question it answers is how busy a place is right now. For venues that opt in, the answer comes from Flux, a small sensor built around a Raspberry Pi 5 and a FLIR Lepton 3.5 thermal camera. A thermal camera records heat instead of light. At 160×120 pixels it cannot recognise a face, and Flux reduces each frame to a count on the device and throws the frame away.

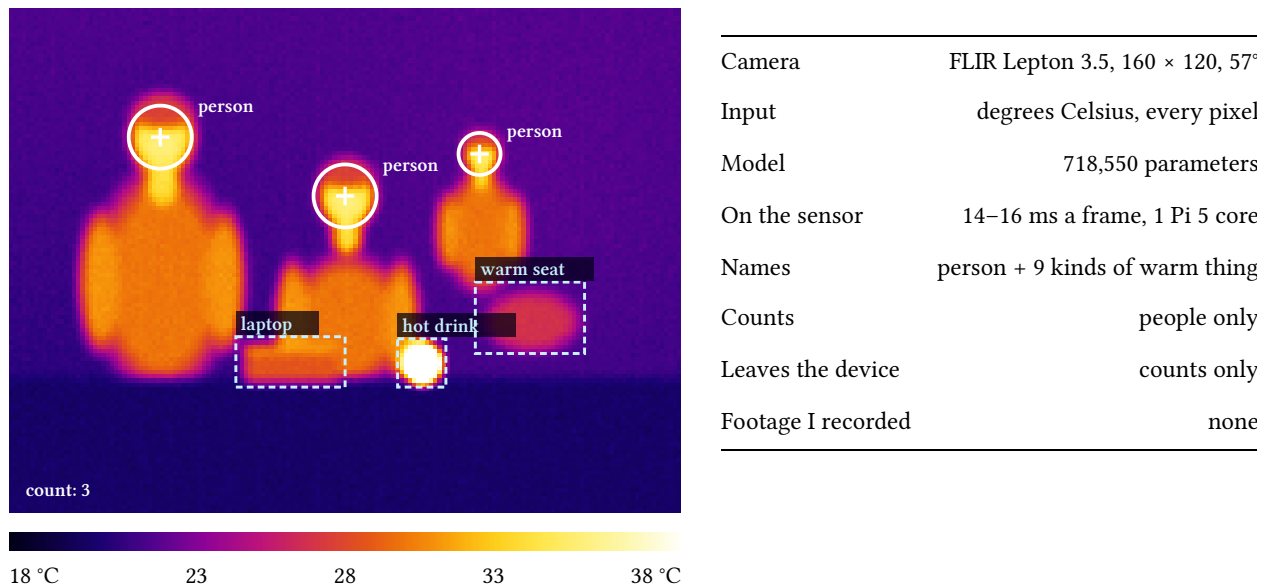


Figure 11. What Owl reads and what it reports. An illustrative 160 × 120 frame, drawn for this paper the way Owl's training generator draws a scene, shown at the camera's true resolution in the iron palette the Lepton uses. Owl puts a point on each person's head and a named box on everything else warm. Only the points are counted: this frame counts as three.

The first counter on Flux was a rule. It found regions warmer than the room, merged them on a coarse grid, and counted the ones large enough to be a person. The rule failed in predictable ways: a hand held up to the lens counted as a person, spread fingers counted as two, a mug of coffee or a laptop could count as somebody, and a packed room merged into one warm mass. Telling these apart is a question of shape, which is what a learned model does well.

Sections 8 to 18 describe Owl, the model that replaced the rule, and everything I learned building it. Four constraints shaped the work from the first day.

1. **It runs on the sensor.** One core of a Raspberry Pi 5, with no GPU and no round trip to a server.
2. **It names kinds of things and nothing more.** No identity, no face features, no tracking from one frame to the next. Only counts leave the device.
3. **I do not record people to train it.** Owl learns from frames I generate and from existing datasets.
4. **Every real frame is licensed.** Each source allows commercial use under its licence, or its authors gave written permission.

My contributions are a generator of radiometric thermal scenes matched to the Lepton 3.5 (Section 11.2); a 718,550-parameter network that finds people as points and names nine other kinds of warm object, decoded on the sensor with physical sanity checks (4.4–4.6); a crowd second look using a density head whose total does not shrink when heads touch (4.6); an evaluation protocol of six held-out real test sets, a 41-case stress test and a per-situation blind-spot report, with the original rule scored on identical frames (5); and fifteen findings, negative ones included, from 24 full training runs (7).

9 Background on thermal people counting

Thermal people counting is usually studied with low-resolution array sensors on the ceiling, such as the 32×24 MLX90640 in PUT Thermo Presence [25], and with higher-resolution cameras for segmentation and pose [24, 26]. Owl borrows from three lines of work.

Objects as points. CenterNet [17] predicts a heat map with a peak at each object's centre and decodes by finding local maxima. Counting then reduces to counting peaks, which is exactly the quantity a venue needs. Its penalty-reduced focal loss, adapted from [18], handles the imbalance between a handful of peak pixels and thousands of background pixels.

Counting by density. Density-based counting [19, 20] regresses a map whose sum is the count. It degrades gracefully when objects overlap, where peak finding merges two touching heads into one.

Learning from generated data. A model trained on generated frames transfers to real ones only as far as the generator covers reality. Domain randomization [21] varies everything the model should ignore so that the real world looks like one more variation. Section 14 shows where that held for Owl and where it did not.

10 The sensor and the baseline

Flux reads the Lepton 3.5 through a PureThermal 3 board over USB. The camera reports radiometric temperatures in degrees Celsius for 160×120 pixels across a 57° horizontal field of view [29]. Through that lens a head three metres away is about 8 pixels wide and a hand 30 cm from the lens about 45.

The baseline rule bins the frame 4×4 , estimates the room as the 20th percentile of the frame, learns fixed warm spots such as radiators per cell over time, keeps cells warm enough to be skin, groups them by eight-connectivity, and counts groups of at least 6 cells, about one head at three metres. Its settings were tuned on a bench with a real person and against constructed scenes, and it works for well-separated people at moderate range. It has no notion of shape, which is the gap Owl fills. Throughout this paper the rule is scored on exactly the same frames as Owl.

11 Owl: method

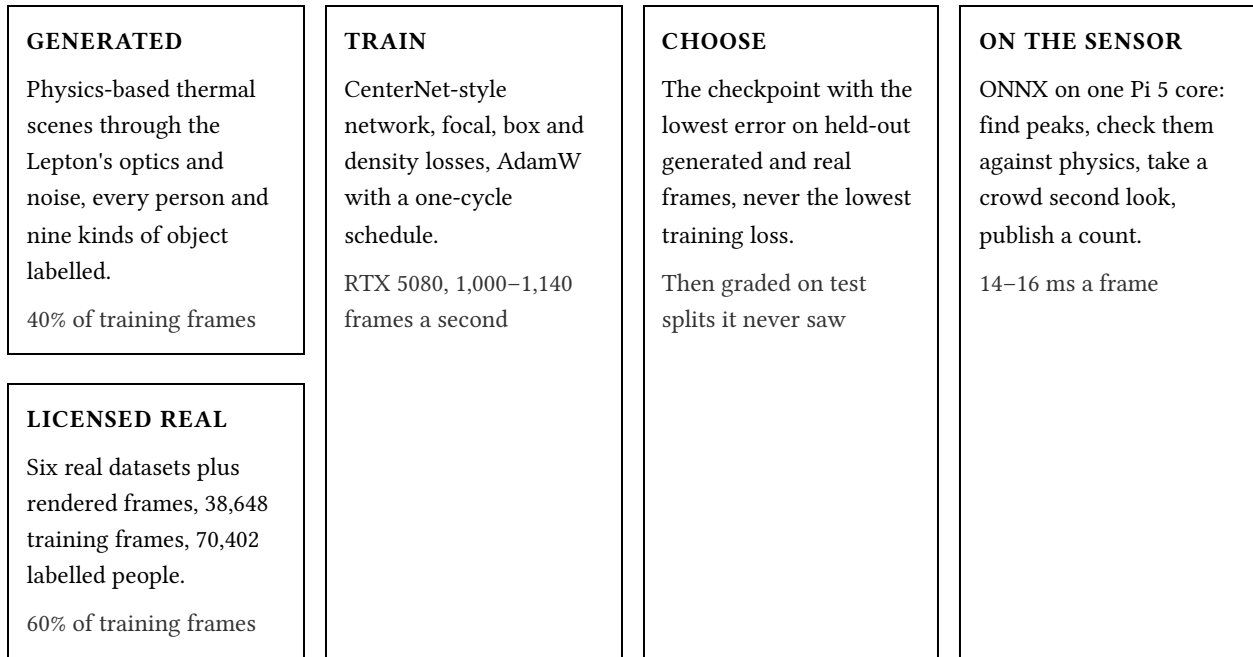


Figure 12. The pipeline. Data on the left, the deployed counter on the right. Model selection uses only validation splits; every number in Section 13 comes from test splits that neither training nor selection touched.

11.1 Formulation

Each frame is labelled with one point per visible person, at the centre of the head, or at the centre of the visible body when the head is out of view. Every other warm object gets a box and one of nine kinds: hand, pet, hot drink, food, laptop, screen, heater, lamp and warm seat. Counting people is finding peaks in the person map. The other nine maps let the screen show why something was not counted, and they give the network somewhere to put the things it must not count.

11.2 A generator matched to the camera

Owl's first version learned only from generated frames, and the generator remained half of every training batch after that. It draws a room at some temperature, people at every distance and pose, and the objects that fool a threshold, at twice the camera's resolution, then averages down, blurs for the lens, and adds the camera's noise. Table 12 lists the physical model, which replaced hand-picked temperatures after the first round of real-footage tests (Finding 4).

Table 12. The physical model behind the generated frames. Values were set from published thermography of skin and clothing.

Surface	Model
Skin	33.4 °C forehead in a 22 °C room, plus 0.14 °C for every degree of room temperature
Clothing	A garment-specific share of the way from the room to the skin; a winter coat reads barely above the room
Hair	3–10 °C above the room
Pets	Coat near room temperature, eyes and ears 35–38 °C
A seat just left	4–8 °C above the room
Rooms	Typical rooms, plus 20% of frames drawn hot at 29–34 °C
Camera	Lens blur, noise, fixed-pattern column stripes, dead pixels, vignetting, several degrees of absolute error, gain error, a crooked mount, the frame after a flat-field correction

Scenes include standing, sitting and lying people, children, hats, hoods, glasses, masks, bald heads, bare torsos, winter coats, people hidden behind tables and bars, one person partly behind another, and crowds. They also include things that must not be counted: hands at the lens, reflections of people in glass and shiny floors, warm footprints, sun on the floor, candles, cold drinks and bulbs hotter than the camera's range.

11.3 Licensed real data

Six real thermal datasets and one set of rendered frames entered training over the project (Table 13). Two are from the Lepton 3.5 itself. Sources that store temperatures (PUT, SDT) are used in degrees Celsius. Sources stored as 8-bit images carry no temperatures, so each frame is mapped onto a plausible room-to-skin range drawn afresh every time it is used (low end 16–28 °C, span 8–16 °C, gamma 0.8–1.25), which stops the network from treating any single mapping as truth. IISER's frames are colour images in the camera's iron palette; I convert them back to one channel by luminance, which rises monotonically along that palette from black through purple, red, orange and yellow to white.

Table 13. Real training data. People are counted in the training split. Licences and permissions are listed in the acknowledgements.

Source	Camera and view	Stored as	Train frames	People	Use
PUT Thermo Presence [25]	MLX90640 32 × 24, ceiling	°C	8,472	24,106	MIT licence
OpenThermalPose2 [26]	Side view, indoor and outdoor	8-bit	8,846	15,849	MIT licence
AAU VAP Trimodal [24]	Wall-mounted, meeting rooms	8-bit	3,964	5,466	CC BY 4.0
SenSys 2021 set [27]	32 × 24 rooms with hot laptops	8-bit	1,239	1,577	CC BY 4.0
SDT, TU Wien [22]	Lepton 3.5 , elevated indoor, one person or none	°C	4,800	3,600	Authors' permission
SDT rendered frames [22]	Rendered to the Lepton's look	°C	8,000	6,000	Authors' permission
Thermal Vision, IISER Bhopal [23]	Lepton 3.5 , classroom of up to 10, lab of up to 5	Iron palette	3,327	13,804	Authors' licence and permission

Real datasets label people and nothing else, so for real frames the loss on the other nine maps is switched off. AAU, SenSys and SDT mark bodies rather than heads; for them the head point is estimated from the box and its target is spread over the pixels the head could occupy ($\sigma = 5$ px), a fix that Finding 7 explains. Test splits are the datasets' own, or, where a dataset has none, contiguous ranges in time: SDT by frame range 60/15/25, and IISER by time within each room 60/15/25 with ten frames dropped at every boundary, because neighbouring frames are near copies.

11.4 Network

The network sees two channels. The first is absolute, $(t - 30) / 8$, so a 60 °C mug can never look like skin. The second is relative to the frame's median, $(t - \text{median}) / 4$, so the several degrees an uncalibrated Lepton reads high or low do not move a person out of range. An encoder halves the resolution four times, to strides 2, 4, 8 and 16 with 24, 48, 72 and 96 channels, using residual blocks [30] with dilation at the deepest stage [31] to widen the view: a hand filling half the frame must be seen as one hand and not five warm blobs. A top-down path fuses the deep features back to stride 4, a 40×30 grid. Three heads read that grid: ten heat maps (bias initialised to -4.6 , so training starts from "nothing here"), a box head that predicts each object's reach in log space, and for the crowd model a density head through a softplus, initialised near empty. The main model has 718,550 parameters and exports to ONNX as a 2.9 MB file.

11.5 Training

The loss is CenterNet's focal loss on the heat maps ($\alpha = 2$, $\beta = 4$), masked by which maps a frame actually labels; an L1 loss on the log of the box reach, at object peaks; and, for the crowd model, squared error per density cell plus a tenth of the error in the total. I train with AdamW (weight decay 10^{-4}), a one-cycle learning-rate schedule with 5% warm-up, gradient clipping at 5, and batches of 64. Fresh runs use a peak rate of 2×10^{-3} ; later versions start from the previous version, keeping every layer whose shape matches, at 2×10^{-4} . Eight CPU workers generate frames on the fly for an RTX 5080. The project ran 24 full training runs totalling 581,700 steps, which is 37.2 million training frames. Each run keeps the checkpoint with the lowest combined error on held-out generated and real validation frames.

11.6 Decoding on the sensor

The sensor runs the ONNX file with ONNX Runtime [13] in full precision on one core. A person is a cell at least as confident as all eight neighbours and above 0.4. Rarer kinds must clear a higher bar before they are named, from 1.25 times the person threshold for hands and hot drinks to 1.75 times for pets, because they are learned from fewer examples. A person peak is dropped when a screen, laptop, heater, lamp or warm seat is more confident in the same 3×3 neighbourhood. Two physical checks run on a 9×9 patch around each person: its warmest pixel must be at least 1°C above the room, taken as the frame's 10th percentile (Finding 3), and its median must not exceed 45°C , which no person reaches. When five or more people are found, a second crowd model reads the same frame, and the published count is the highest of the main model's peaks, the crowd model's peaks and the rounded density total.

12 Owl: evaluation protocol

Real test sets. Six test splits that no model trained or was selected on: Flux's camera model in indoor rooms (SDT, 2,000 frames), Flux's camera model in a crowded classroom and lab (IISER, 1,375 frames, 4,882 people), a ceiling camera (2,347 frames, 8,043 people), rooms with hot laptops (265 frames), wall-mounted meeting rooms (813 frames, 1,061 people) and a side view (2,151 frames, 3,872 people). A frame is *exact* when the count equals the truth and *within one* when it is off by at most one person. The rule is scored on the same frames.

Stress test. 41 hard situations, each generated on purpose 150 times from a fixed seed and scored alone, counted exactly as the sensor counts, crowd model included.

Blind-spot report. Every real test set split by situation (by pose for SDT and by number of people elsewhere), and every stress case. Anything under 90% exact or 97% within one is flagged.

13 Owl: results

Figure 13 compares the shipped pair, owl-4.9 and its crowd model, with the rule on all six real test sets. On five of them Owl counts 82–96% of frames exactly and is within one person on 96–100%. On the crowded classroom set, the hardest test in this paper, it is exact on 39% and within one on 74%, against the rule's 33% exact.

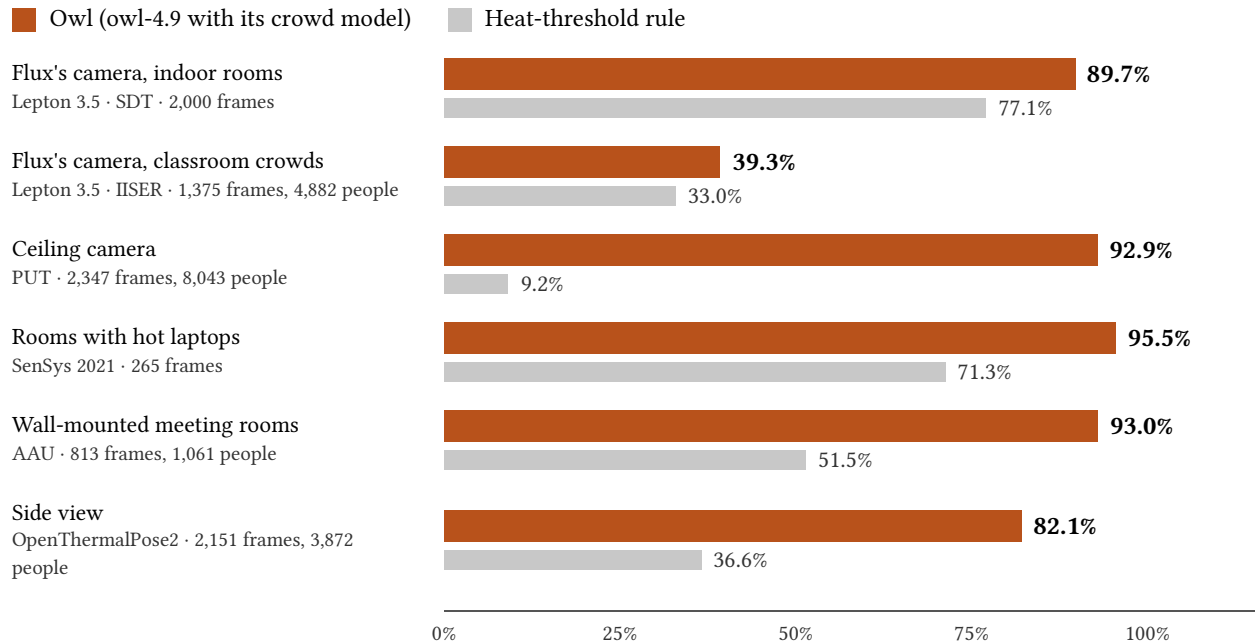


Figure 13. Frames counted exactly right on six real test sets that no model trained or was selected on. The first two sets come from Flux's own camera model, the FLIR Lepton 3.5. The rule is scored on identical frames.

13.1 From owl-1 to owl-4.9

Owl went through sixteen named versions in four days. Figure 14 follows the five original real test sets across the versions that shipped. The jump between owl-2 and owl-3 is the first finding of this paper: the same network with real frames mixed in. The dip on the meeting rooms at owl-4.7 and its recovery at owl-4.8 is a data-mix trade-off that Figure 15 isolates.

■ Owl, exact ---- the rule Missing points were not measured for that version.

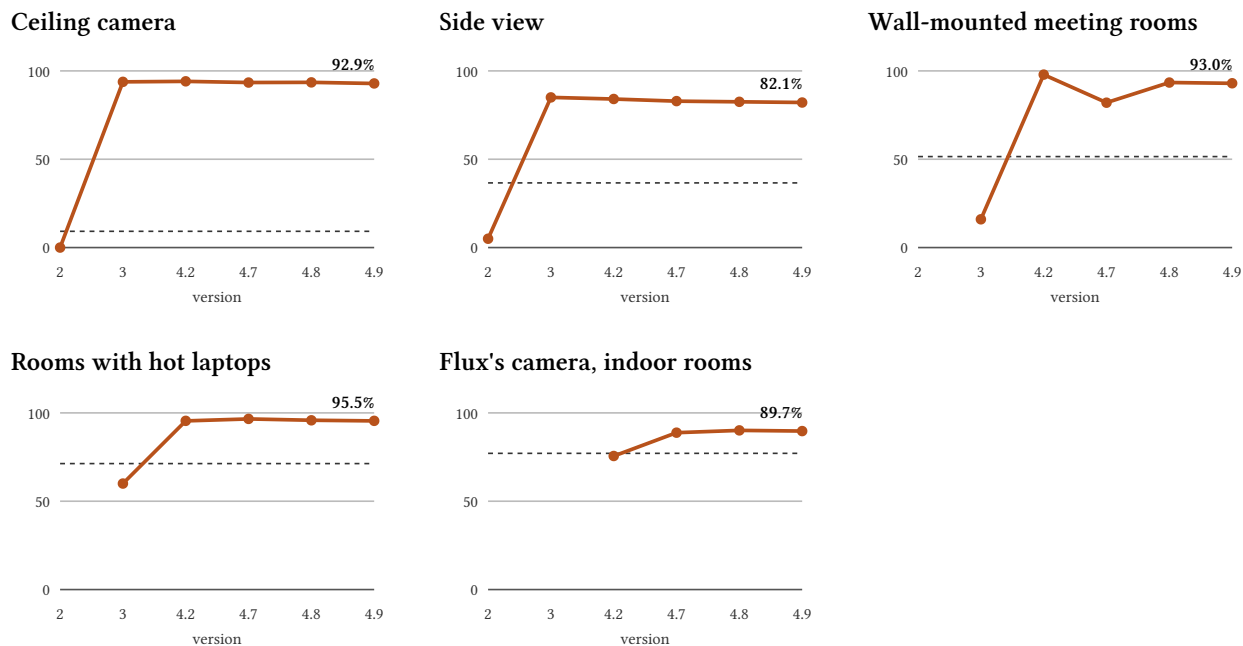


Figure 14. Exact counts on each real test set by version. owl-2 learned only from generated frames; owl-3 added real frames; owl-4.2 was the first shipped with meeting rooms, laptop rooms and measured physics; owl-4.7 and owl-4.8 added frames from the Lepton 3.5; owl-4.9 added real classroom crowds. owl-3's side-view figure includes the background fix of Finding 3 (74% before it).

13.2 Like-for-like comparison of the last two versions

When owl-4.9 was trained I re-graded owl-4.8 and owl-4.9 on the full test splits with identical code (Table 14). On the five earlier sets nothing moved by more than 0.6 points, which is within the sampling noise of sets this size. On the classroom crowds the error fell by more than three quarters.

Table 14. owl-4.8 and owl-4.9, each with its crowd model, graded together on full test splits. Exact / within one, in percent.

Test set	Rule exact	owl-4.8	owl-4.9
Flux's camera, indoor rooms	77.1	90.1 / 100	89.7 / 100
Flux's camera, classroom crowds	33.0	11.5 / 20.6	39.3 / 73.9
Ceiling camera	9.2	93.5 / 99.7	92.9 / 99.7
Rooms with hot laptops	71.3	95.8 / 100	95.5 / 100
Wall-mounted meeting rooms	51.5	93.4 / 99.9	93.0 / 99.1
Side view	36.6	82.5 / 96.6	82.1 / 96.3
41-case stress test, mean exact	—	94.5	94.9

13.3 Crowds on Flux's own camera

The IISER classroom set is the only real crowd I have from the Lepton 3.5: rows of up to ten people, each row partly hidden behind the one in front. owl-4.8 had never seen anything like it and overcounted badly, averaging 8.1 people on frames that held 3.6. Figure 16 separates the two halves of the fix. Training the main model on the classroom frames lifted exact counts from 11.5% to 28.6%. Retraining the crowd model as well took them to 39.3%, and in rooms of six to ten people moved within-one from 3.4% to 96.1%.

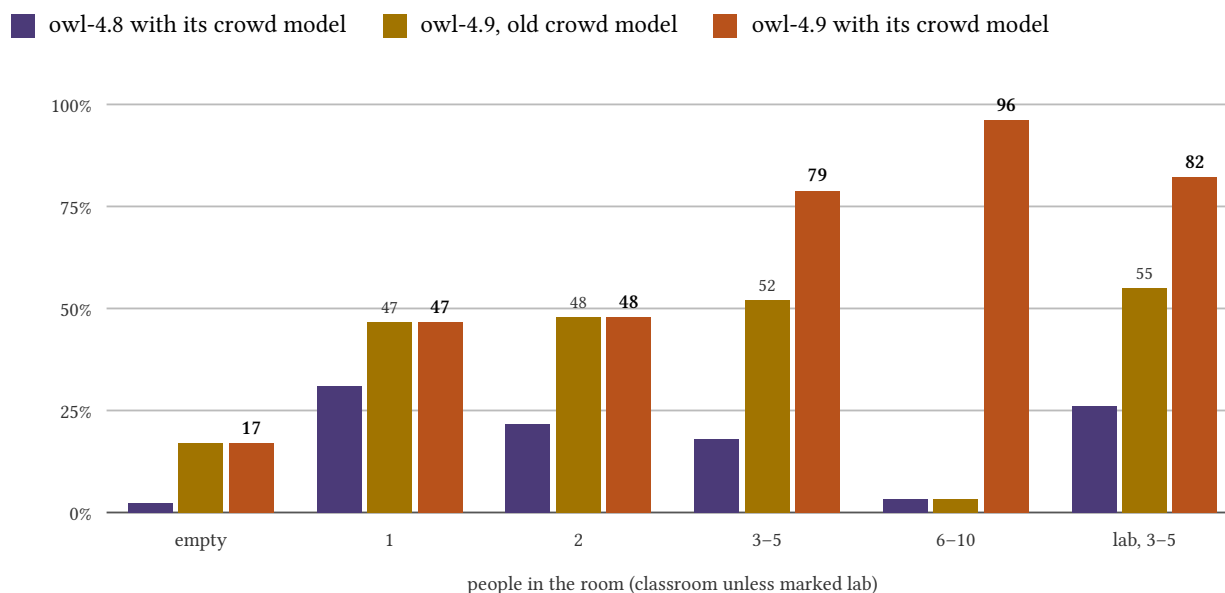


Figure 16. Frames within one person of the truth on IISER's held-out classroom and lab frames, by how many people were in the room. The crowd model's contribution is the difference between the gold and orange bars; it decides the six-to-ten bucket almost on its own.

When the crowd model steps in is a single setting. Handing over earlier was tested on the same frames and was worse on both the classroom crowds and the stress test (Table 15), so the sensor keeps five.

Table 15. When to ask the crowd model, owl-4.9 pair.

Crowd model asked from	Classroom exact	Within one	Stress mean	Blind spots
3 people	36.9	72.3	94.3	12
4 people	37.5	73.1	94.6	10
5 people (shipped)	39.3	73.9	94.9	11

13.4 Hard cases

Figure 17 shows all 41 stress cases for the shipped model. Everything that fooled the rule by design (a hand at the lens, a hot mug, a monitor, a radiator, a pet, a reflection, warm footprints, a candle) is right 96–100% of the time. Every remaining weakness is about people hidden by people or lying flat, and in

all of them the count is still within one person on 85% of frames or more.

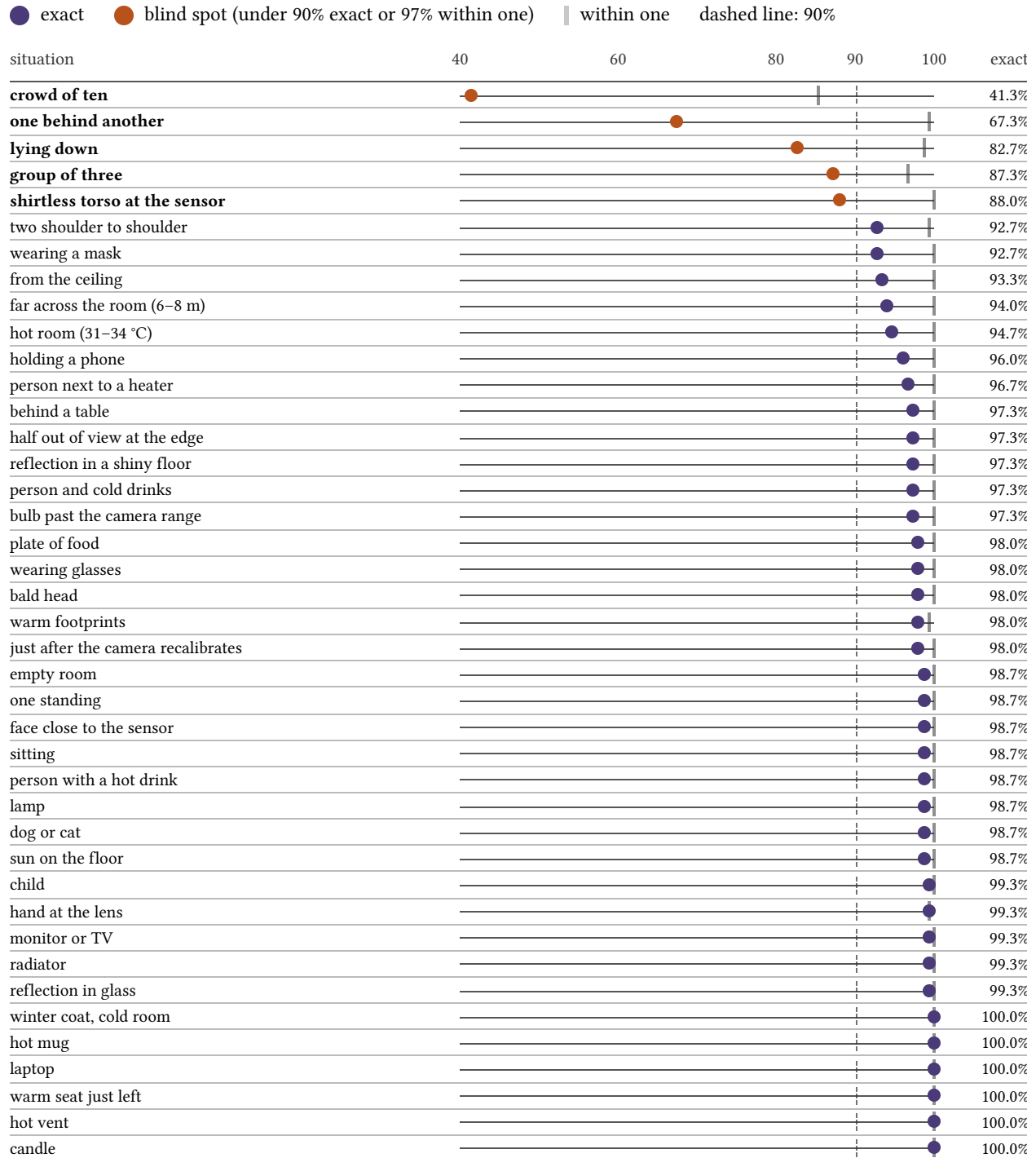


Figure 17. The 41-case stress test for owl-4.9 with its crowd model, 150 generated frames per case, sorted from hardest to easiest. Mean 94.9% exact and 99.5% within one.

13.5 Naming everything else

On 3,000 held-out generated frames, owl-4.9 finds the nine other kinds as Table 16 shows. These names are shown on the sensor's screen to explain why something was not counted; they are never counted and never sent.

Table 16. Object naming on held-out generated frames. Found: share of real objects with a peak. Right when named: share of named peaks that were that kind.

Kind	Objects	Found	Right when named
person	10,461	74.9%	89.7%
hand	682	69.5%	84.3%
pet	617	67.1%	92.4%
hot drink	836	84.2%	83.0%
food	441	81.2%	91.6%
laptop	768	68.2%	79.8%
screen	491	60.5%	80.7%
heater	794	78.7%	85.5%
lamp	316	77.5%	79.0%
warm seat	586	58.7%	83.1%

The same 3,000 frames are deliberately harder than real rooms, with crowds of up to ten drawn often. On them owl-4.9 counts 59.3% exactly and 84.3% within one, against the rule's 20.2% and 49.9%; empty rooms are right 92.3% of the time and single people 80.1%. Because the generator grew harder with each version, these synthetic figures are not comparable across versions, which is one reason every decision in this paper rests on the real test sets.

13.6 Speed and size

On the sensor, owl-4.8 and owl-4.9 take 14–16 ms per frame on one core of the Pi 5, measured by the sensor's own self test on its own camera. That is fast enough to read every frame the camera sends. An 8-bit quantised export runs 2.1 times faster at a quarter of the size, but it cost accuracy (Finding 11), so the sensor runs full precision.

14 Owl: findings

Every change to Owl was kept or thrown out on real held-out footage. These are the results that shaped it, in the order I found them, including the ideas that failed.

F1 *Generated frames alone fail on real footage*

owl-2 counted 80% of generated frames exactly and found 73–94% of the warm objects. On real footage it counted 0% of the ceiling frames and 5% of the side-view frames exactly. It had only seen clean drawn shapes and read the texture of a real room as people.

F2 *A little real data fixes most of it*

A 500-step test with real frames mixed in went from 0% to 81% exact on the real ceiling set. The full run, owl-3, reached 94% on the ceiling set and 74% on the side view. From then on, generated frames also gained real-looking texture and low-resolution blocking.

F3 *Judge warmth against the cool end of the frame*

The physical check compared each person with the room, taken as the frame's median. When a person fills the view, the median is the person, and the check threw out the person the model had just found. Taking the room as the 10th percentile lifted the real side view from 74% to 85% exact and 86% to 98% within one, hot rooms from 47% to 68%, and a bare torso at the sensor from 16% to 30%, with the ceiling set and every false-alarm case unchanged.

F4 *Physics beats tricks*

Two augmentations that seemed sensible, skewing the absolute-temperature channel and drawing real group frames three times as often, changed nothing on the ceiling and side-view sets and cost a few points on unseen rooms. Replacing hand-picked temperatures with measured skin and clothing physics lifted the unseen meeting rooms from 24% to 33% exact and from 58% to 85% within one.

F5 *A bigger network did not help*

Widening the network from 24 to 32 base channels tied the smaller one (94.2% and 73.9% against 93.8% and 74.2% on the ceiling and side view). Accuracy was limited by data, which set the direction for everything after.

F6 *More of a hard case can cost more than it gains*

owl-4.4 drew more people lying down, more bare torsos and more bodies at the sensor, the two single-person cases still under 90%. It lost the real meeting rooms, falling from 97.9% to 60.9% exact, with no measurable gain on the cases it targeted. The change was reverted.

F7 *A sharp target at a guessed point teaches the model to miss*

A finer 80 × 60 output grid (owl-5) keeps heads a few pixels apart as two people. But on the datasets whose head points are estimated from a body box, a sharp target at the guess taught it to miss people: the meeting rooms fell from 97.9% to 67.2%. Spreading those targets over the pixels the head could occupy fixed the cause, and became standard for every later version.

F8 *Density counts crowds that peaks merge*

Used as a second look in crowds, the finer-grid model lifted a crowd of ten from 41% to 63% within one. A density head, whose total does not fall when heads touch, lifted it to 82% and put one person behind another at 100% within one, with all four real test sets unchanged. It became the crowd model (owl-4.3).

F9 *Data shares trade off, and the right data ends the trade*

Adding real Lepton frames (SDT) improved Flux's own camera but every way of making room for them hurt the meeting rooms: cutting their share to 15% cost 25 points, and owl-4.6 gained 13 points on the Lepton while losing 14 on the meeting rooms. owl-4.7 chose the camera Flux uses (88.8% against owl-4.2's 75.6%). Adding SDT's rendered Lepton-style frames in owl-4.8 improved both at once (Figure 15).

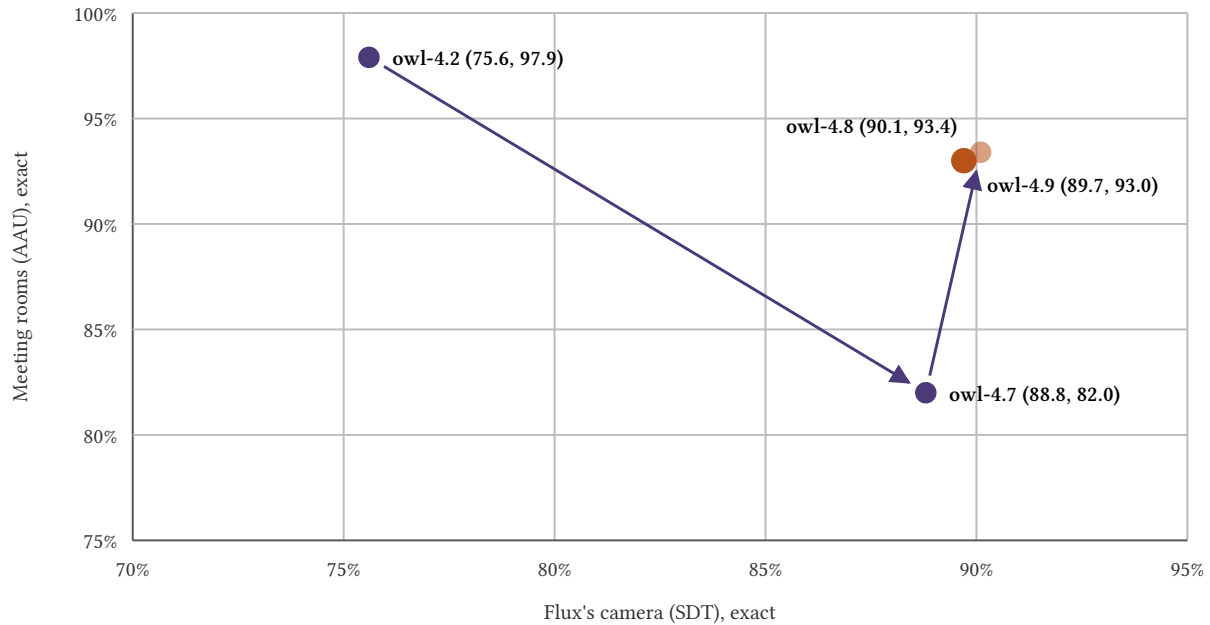


Figure 15. The trade between Flux's own camera (SDT, horizontal) and the wall-mounted meeting rooms (AAU, vertical). owl-4.7 bought the camera with the meeting rooms; owl-4.8, trained with rendered Lepton frames as well, recovered the meeting rooms without giving the camera back. owl-4.9 sits within noise of owl-4.8.

F10 *Validation must never be reweighted*

When training drew group frames three times as often, the validation draw reused that weighting by accident, and owl-3.2 looked worse than owl-3 on what was really a harder mix. Validation frames are now drawn as they come, so runs stay comparable.

F11 *Quantising cost more than it saved*

An 8-bit export ran 2.1 times faster at a quarter of the size, but cost owl-3 eight points on the real ceiling set (94% to 86%), and 84% even when calibrated on real frames. At 14–16 ms per frame in full precision, speed was never the constraint, so the sensor keeps full precision.

F12 *A newer base can make a worse crowd model*

Retraining the crowd model starting from owl-4.8 instead of its own earlier weights dropped a crowd of ten to 18.7% exact. The older crowd model stayed until real crowd data arrived.

F13 *A real room shows what no test set does*

On the sensor in a real room, owl-3 named a pet in a room with none, sometimes counted a monitor, and missed a shirtless torso close to the camera. Three changes answered them: higher naming bars for rarer kinds, dropping a person where a screen-like object is more confident, and bare skin and bodies filling the view in the generated frames. On the stress test a shirtless torso at the sensor went from 16% under owl-3 to 88% under owl-4.9.

F14 *Real crowds on the real camera were the missing data*

After owl-4.8, training on the same data had stopped helping. Only after IISER Bhopal licensed its Lepton 3.5 classroom crowds to me did the crowd numbers move: 11.5% to 39.3% exact, an average error of 4.66 people to 1.03, and rooms of six to ten from 3.4% to 96.1% within one, while no other test set moved by more than 0.6 points.

F15 *The model has learned what this data can teach*

owl-4.9's real validation score stopped rising about a quarter of the way through its run (Figure 18). Longer training would only move sideways. The next gain needs new data, specifically more real crowds and side views from this camera.



Figure 18. owl-4.9's real validation score every 500 steps over its 20,000-step run. Both curves are flat after about 5,000 steps. The validation mix includes the new classroom crowds, so these values are lower than earlier versions' and are not comparable with them.

15 Privacy by design

Flux exists to tell people how busy a place is. It must never become a camera that watches them, and Owl was built so that it cannot.

- **Kinds, never who.** Owl's outputs are ten kinds of thing. It has no identity output, no face class and no face features. IISER labels faces, and I use each face box only as the point where a head is;

nothing in Owl learns what a face looks like.

- **No memory between frames.** Each frame is read on its own and forgotten. Nothing follows a person from one frame to the next.
- **Counts are all that leave.** The network's points and boxes exist only in memory. Boxes are drawn only on units with a screen and are never sent. What leaves the sensor is a handful of numbers.
- **Enforced by tests.** Tests in the sensor's suite fail the build if it gains identity or tracking, carries anything between frames, or sends names off the device.
- **Data used as licensed.** Every dataset is used as its licence or permission allows. None of their images are redistributed or published, and none can be reconstructed from the model. IISER's licence also forbids using its data to identify anyone, which Owl cannot do.

The model files are kept private, like Flock's crowd-forecasting model. The generator, network and training and grading code are public. Without a model on disk the sensor counts with the rule, and says so.

16 Limitations of the counter

- **People hidden by people.** A side view of three to five people is exact on 40% of frames (85.5% within one), and one person behind another on 67% (99% within one). No single frame shows a person who is fully hidden.
- **Lying and sitting on the Lepton.** On SDT, lying people are exact on 81% of frames and sitting people 83%, though both are always within one.
- **The colour-image domain gap.** IISER's frames carry the camera's automatic contrast instead of temperatures, so a seat someone just left is stretched to the same brightness as skin. On those frames Owl still overcounts empty and nearly empty rooms. Flux reads temperatures, where a warm seat is several degrees cooler than a face, so I expect the sensor to do better, but I have not measured it in a venue.
- **No venue ground truth yet.** All six test sets come from other people's rooms. A study in a working venue, with counts taken by hand, has not been done.
- **The stress test is generated.** Its cases come from the same family of generator as half the training data, so it measures coverage of known hard cases and can flatter the model on them.
- **Single-person Lepton data.** SDT frames hold one person or none, so Flux's camera at moderate group sizes is tested mainly through IISER's colour frames.

17 Future work on the counter

The clearest next step is data. A request to use the IPHD thermal dataset is outstanding, and the gap that matters most is real radiometric crowds from the Lepton 3.5 in venues. A study in a working venue, with hand counts as ground truth, would replace the stress test's estimate of real-world accuracy with a measurement. Flux also carries a doorway counter, an 8×8 time-of-flight sensor that tells

arrivals from departures. Combining its running total with Owl's count of people in view, and estimating dwell time with Little's law [28], can answer how many people are inside rather than how many are visible, without identifying anyone. Finally, each crowd decision currently rests on one frame; agreeing a count across consecutive frames, without following individuals, may steady crowded readings further.

18 Conclusions on the counter

Owl shows that a model small enough for one core of a Raspberry Pi can count people on a 160×120 thermal camera far better than a heat threshold, without anyone being recorded to train it and without the sensor ever keeping a frame. Generated frames gave it a vocabulary of warm things that the rule confused with people; licensed real frames taught it what real rooms look like; and each real dataset, once added, closed the gap it was chosen for. Across six real test sets, owl-4.9 counts five of them exactly 82–96% of the time where the rule managed 9–77%, and on real classroom crowds it misses by one person on average where its predecessor missed by almost five. The work also showed where the next improvement has to come from: not more training, but more real footage of crowds from the camera Flux uses.

19 Joining the forecast and the counter

Both studies serve one number in Flock: how busy a place will be when a group arrives. Each leaves open something the other can supply. This section proposes work that has not been run, and reports no results.

19.1 What each study leaves open

The forecast predicts busyness normalised per venue, so that 100 is the busiest that venue gets (Section 4.1). That is a relative scale. A score of 70 says nothing about how many people are inside, and the same score could mean a dozen people in a small bar or two hundred in a large one. Its live readings arrive at most once an hour, and the freshness of the latest one is worth a great deal: a venue's departure from its curve fades within about three hours (Section 3.2).

Owl counts people on the device and sends only the count. It has been measured on licensed real test sets, not yet in a working venue, and it undercounts people hidden behind others in dense crowds (Sections 16 and 17).

19.2 Counts as ground truth

A Flux unit in a venue that Flock already reads would give that venue an absolute scale. Its headcount, paired with the vendor's live score for the same hours, would show how many people a given score means at that venue, and whether the five crowd levels match what a group sees at the door. Flux's doorway counter adds arrivals and departures, so the sensor can estimate how many people are inside rather than how many are in view (Section 17).

It would also give the nowcast a fresher reading. The nowcast weights the latest reading by its age (Section 4.4). A sensor that reports every 30 seconds would make that reading minutes old instead of up to an hour.

19.3 What would be measured first

- One unit, installed with the owner's agreement, at a venue the hourly collector already reads.
- Hand counts at the door on a few busy nights, to score Owl in that room before its counts are used as ground truth.
- Hour by hour over several weeks: the vendor's live score against Flux's count, the score per person, and how often the crowd level names what the count says.
- The October retrain of Section 6.4, scored exactly as pre-registered. The counts are a separate test and change none of its rules.

None of this has been run yet. It is the plan that would turn the two studies into one measurement.

Acknowledgements

Owl would not count real rooms without the people who shared their data. The Computer Vision Lab at TU Wien (Christopher Pramerdorfer, Julian Strohmayer and Martin Kampel) granted permission to use the SDT dataset, real and rendered. Arijit Samal and Haroon R. Lone of IISER Bhopal licensed the Thermal Vision dataset and granted permission to use it. PUT Thermo Presence (PUT Vision Lab) and OpenThermalPose2 (ISSAI) are used under the MIT licence; the AAU VAP Trimodal dataset (Aalborg University) and the SenSys 2021 low-resolution set are used under Creative Commons Attribution 4.0. None of their images are included in Flux or in this paper.

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Appendix A: Generalisation across cities

Leave-one-city-out cross-validation gives R^2 0.653 on all rows, and the geographic holdout 0.772 (Figure A1). These blended figures include the weekly anchor rows, whose correct delta is zero, and are not accuracy claims for the app. On these blended rows the held-out cities score better than cross-validation. On the rows the app serves the order reverses, R^2 0.127 out of fold against 0.040 on the held-out cities (Section 5.2): moving to a city the model has never seen costs accuracy, even though the gain over the curve holds in each of the three. The four largest cities and the smallest are the hardest.

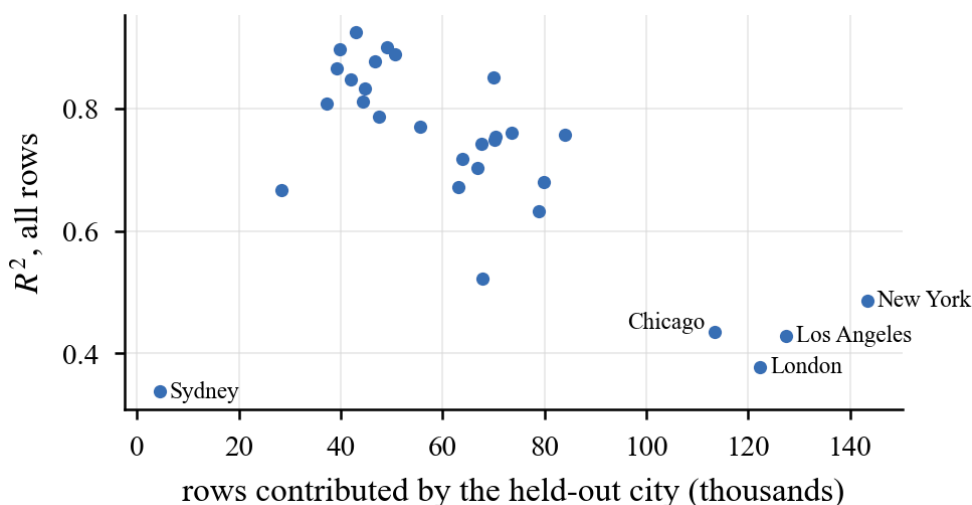


Figure A1. Leave-one-city-out R^2 per held-out city on all rows (blended; not an accuracy claim), against the rows the city contributes.

Appendix B: Choices made after seeing data

Hypotheses H1 to H3 were written after the preliminary findings of Section 3 and several of the analyses in Section 5 had been run; Section 6.4 is the first test fixed in advance. The persistence measurements of Section 3.2 cover 2026-09-01 to 09-08 and so overlap the three test days; they informed the decision to add a nowcast, while its weights were fitted on 09-01 to 09-05 only. The serving clamp, the rail pull, the display map, the offset weight and every nowcast weight were chosen on a split earlier than the rows that report them. Weekly rows carried the weather of their collection moment, a leak cleared from 3,246,456 rows. Of rows recording “no event nearby”, 56.1% were schema defaults correlated with city, which is why Section 5.4 disables the events lookup for both arms.